

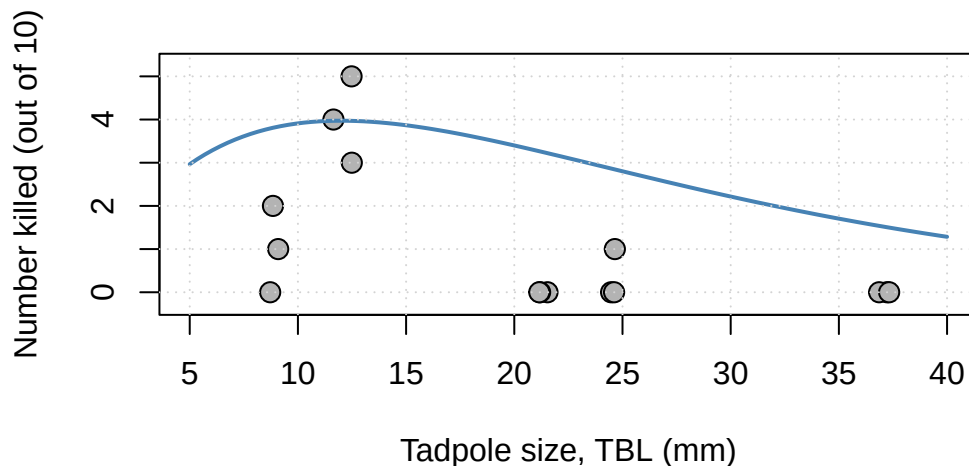
Lab 1, Act 3: Defend

Does the evidence actually support our candidate?

NRES 746 – name: _____ group: _____

Picking up where we left off

In Act 1, your group diagnosed the tadpole data as hump-shaped and picked the Ricker function as your top candidate. In Act 2, you put specific numbers on it ($a = 0.9$, $b = 1/12$) and derived its peak, $x^* = 12$ mm. Today we put that candidate on trial: does it actually predict the real data well, or does it just *look* plausible?



Part 1: Does the shape earn its keep? (individual, ~10 min)

A model earns its keep by predicting better than a “dumb” baseline. One simple way to score any model: for each data point, compute the **residual** (observed – predicted), square it (so overshoots and undershoots both count against the model, and big misses count extra – this is the same squared-deviation idea from Act 1’s variance calculation), and add up the squares. Smaller total = better fit. This total is often called the **sum of squared residuals (SSR)**.

Model 1: our Ricker candidate. Using $a = 0.9$, $b = 1/12$ from Act 2, fill in the missing residuals:

TBL (mm)	Observed mean	Ricker predicted	Residual	Squared residual
9	1.00	3.83		8.01
12	4.00	3.97		0.00
21	0.00	3.28		10.76
25	0.33	2.80		6.10
37	0.00	1.53		2.34

TBL (mm)	Observed mean	Ricker predicted	Residual	Squared residual
			SSR:	27.21

Model 2: the “null” model. The simplest possible model: ignore tadpole size entirely, and always predict the overall average. First, find that average (using either Act 1’s data table or the mean of the five group means):

$\bar{y} =$ _____

Now fill in the rest of the table yourself:

TBL (mm)	Observed mean	Null predicted	Residual	Squared residual
9	1.00	_____	_____	_____
12	4.00	_____	_____	_____
21	0.00	_____	_____	_____
25	0.33	_____	_____	_____
37	0.00	_____	_____	_____
			SSR:	_____

Compare. Which model has the smaller (better) SSR – the Ricker curve, or the null model that ignores body size entirely? Is this what you expected, given Acts 1 and 2?

What does today’s result mean for the tadpole predation model (predation as a function of size)? What do you take away from the above result?

Group discussion (~10 min)

Compare your work with your group:

- Compare your SSR(null) calculations. Did everyone get the same total? Resolve any arithmetic disagreements.
- Debate about whether you think the null model is the best you can do. Where does your group land? What would you try next?
- Look back at Act 1’s variance table – most groups had only 3 replicate tanks. Given that, how much would you trust any single group’s mean? Would you be surprised if a different set of tanks gave noticeably different numbers, especially for TBL=25?
- As a group: are you ready to *defend* the Ricker function as the right model for this system, or do you want to reserve judgment? What (if anything) would change your mind?

Revise and reflect (individual, ~8 min)

Revise any of your answers above based on the group discussion. Then answer:

1. In your own words: what's the difference between a model having the *wrong shape* versus the *right shape but poorly-chosen parameters*? Based on today's evidence, which one is the bigger problem for our Ricker candidate?

2. If you could plug in the *best possible* a and b for the Ricker shape – not just eyeballed guesses – do you think it could beat the null model? Why or why not?

3. Muddiest point: what's the one thing from this part of the lab you'd like to see explained again?
